Using LiDAR to Understand Ecosystems

Badge Awarded to

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LiDAR (Light Detection and Ranging) is an optical remote sensing technology that uses laser pulses to determine the distance between the sensor and a surface or object. LiDAR has emerged as one of the most important sources of data for topographic mapping, vegetation analysis, and 3D modeling of infrastructure. Specific uses of this technology include floodplain mapping, transportation planning and design, resource and environmental management, and emergency response. New uses for LiDAR data are emerging. An understanding of LiDAR technology and its application in GIS will assist watershed and conservation professionals to be on the leading edge of data driven adaptations for the changing natural environment.

This microcredential is an introduction to the capabilities of LiDAR and associated technologies to support watershed resilience. Learners will use data sources in a broad range of applications to understand ecosystems and to build watershed resilience.

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Issuer

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Holland College is the provincial community college for the Canadian province of Prince Edward Island (PEI).

Criteria

The assessment for this microcredential will require learners to:

-select appropriate tools and data types to include in the plan to support watershed resilience.

-gather LiDAR data, satellite imagery, drone imagery, and other sources.

-interpret data using GIS software.

-justify decisions regarding tools and data types.

In order to develop competency, learners will:

-understand the fundamentals of LiDAR, drone and satellite imagery

-understand the application of data to watershed management

-examine how LiDAR and associated data can be collected for a project

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